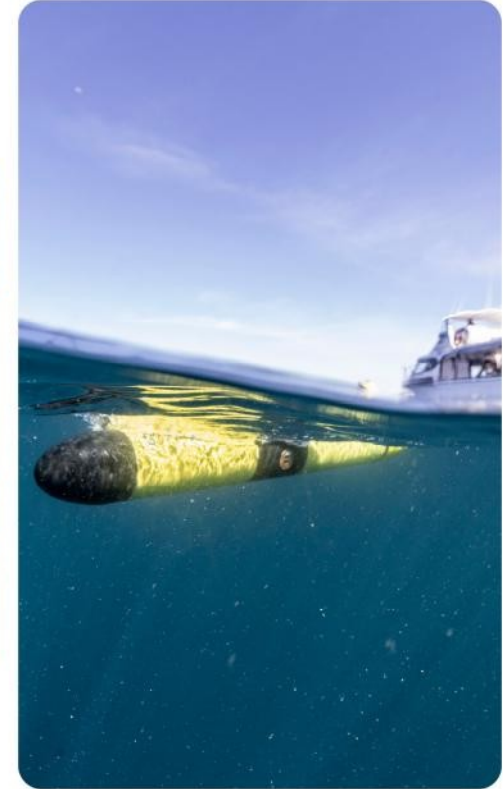


AODN Cloud Optimised

Fast and efficient access to huge datasets

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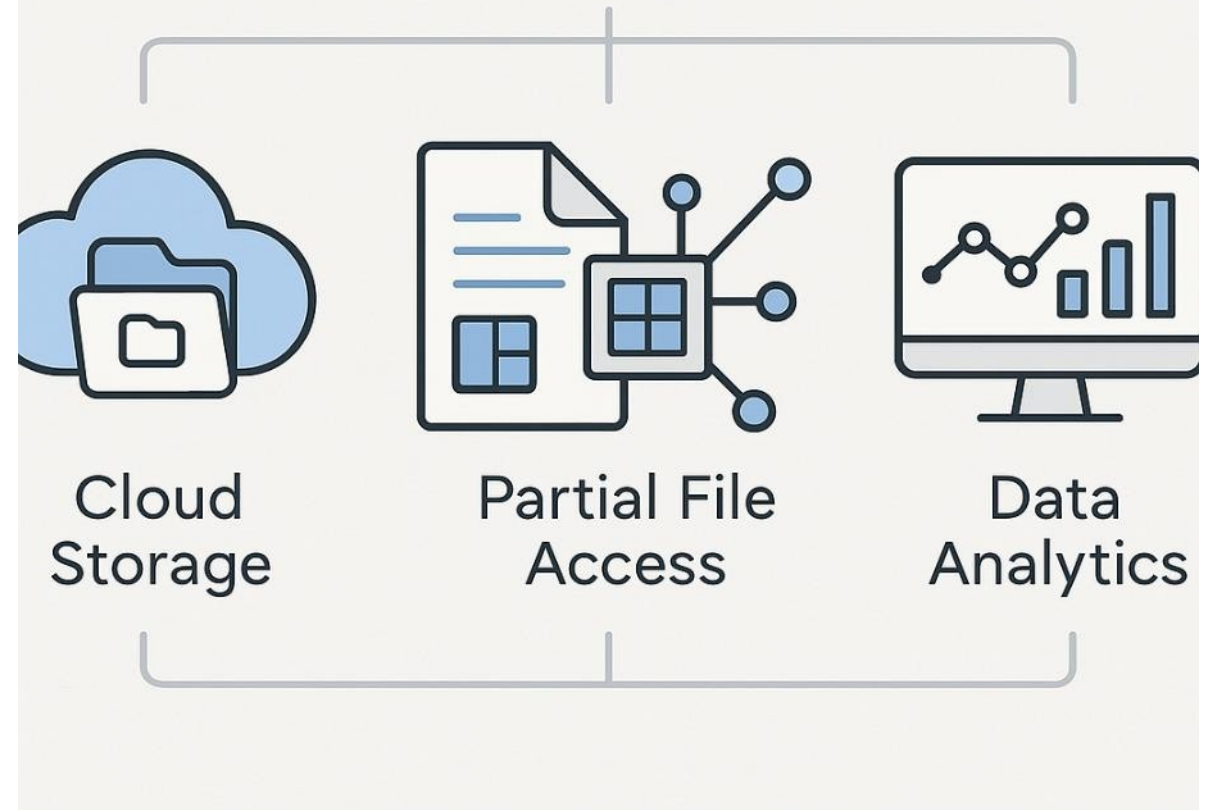
IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

What is a cloud optimised format?

- A file format designed for **efficient use** in cloud environments.
- Enables **fast, scalable access** to large datasets over the internet.
- Supports **partial reads and parallel processing**.
- Ideal for **big data analytics**, machine learning, and remote sensing.

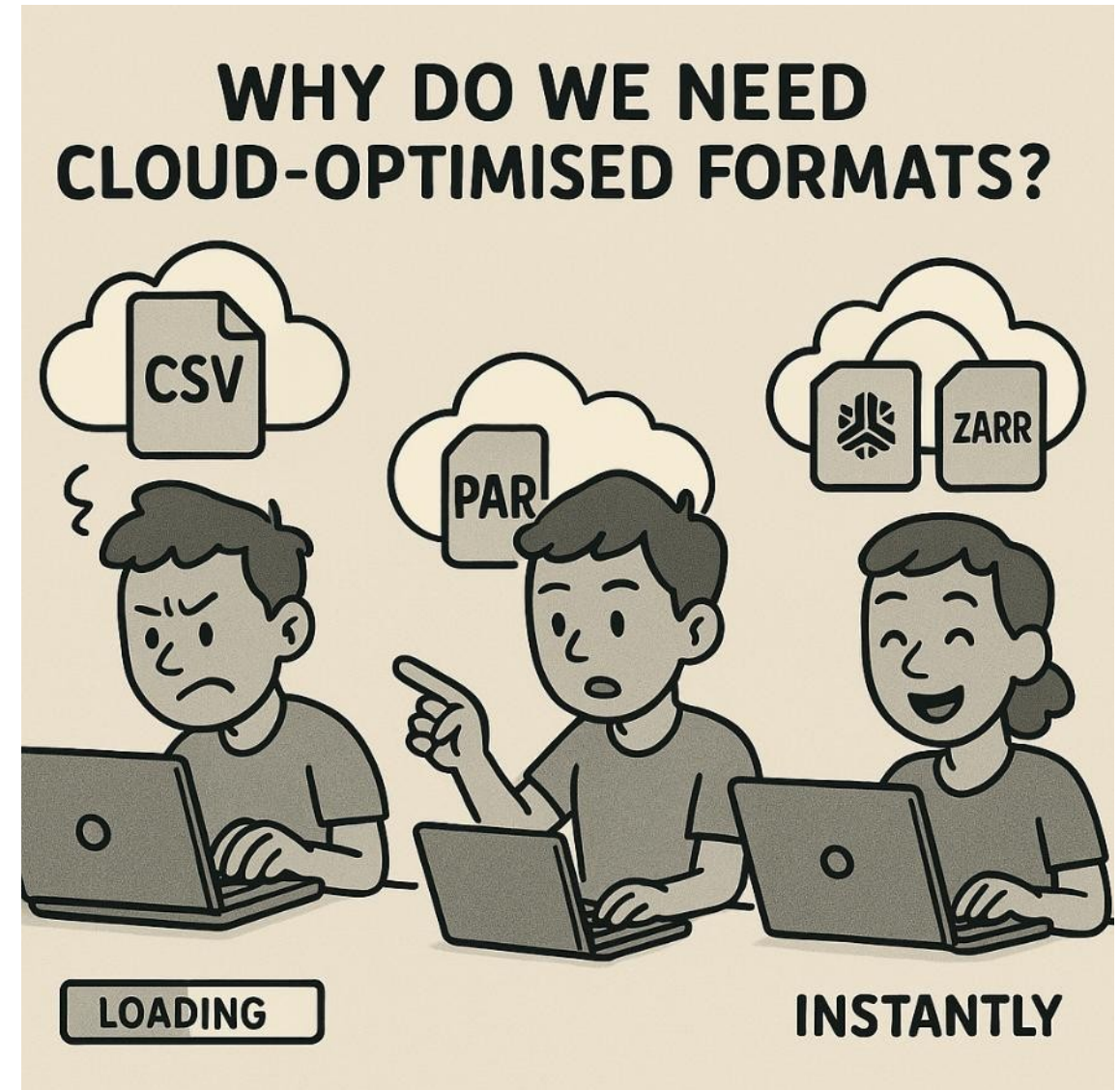
Have you ever used a cloud optimised file?

CLOUD-OPTIMISED FILE FORMAT



Why Do We Need Cloud-Optimised Formats?

- **Explosion of data:** Modern applications generate massive datasets—traditional formats struggle to keep up.
- **Remote access:** Cloud environments require formats that support partial reads and efficient streaming.
- **Scalability:** Cloud-optimised formats enable distributed processing across multiple nodes.
- **Cost efficiency:** Compression and selective access reduce storage and bandwidth costs.
- **Interoperability:** These formats integrate seamlessly with cloud-native tools and platforms.
- **Performance:** Faster queries and analytics with columnar and chunked storage structures.



Characteristics of Cloud-Optimised Formats

- **Chunked or columnar storage** for efficient querying.
- **Built-in compression** to reduce storage and bandwidth.
- **Self-describing metadata** for schema evolution and validation.
- Support for **parallel reads/writes** in distributed systems.
- **Compatibility** with cloud-native tools (e.g., AWS Athena, BigQuery).

Key Characteristics of Cloud-Optimised File Formats



COLUMNAR
STORAGE



COMPRESSION



METADATA



PARALLEL
ACCESS



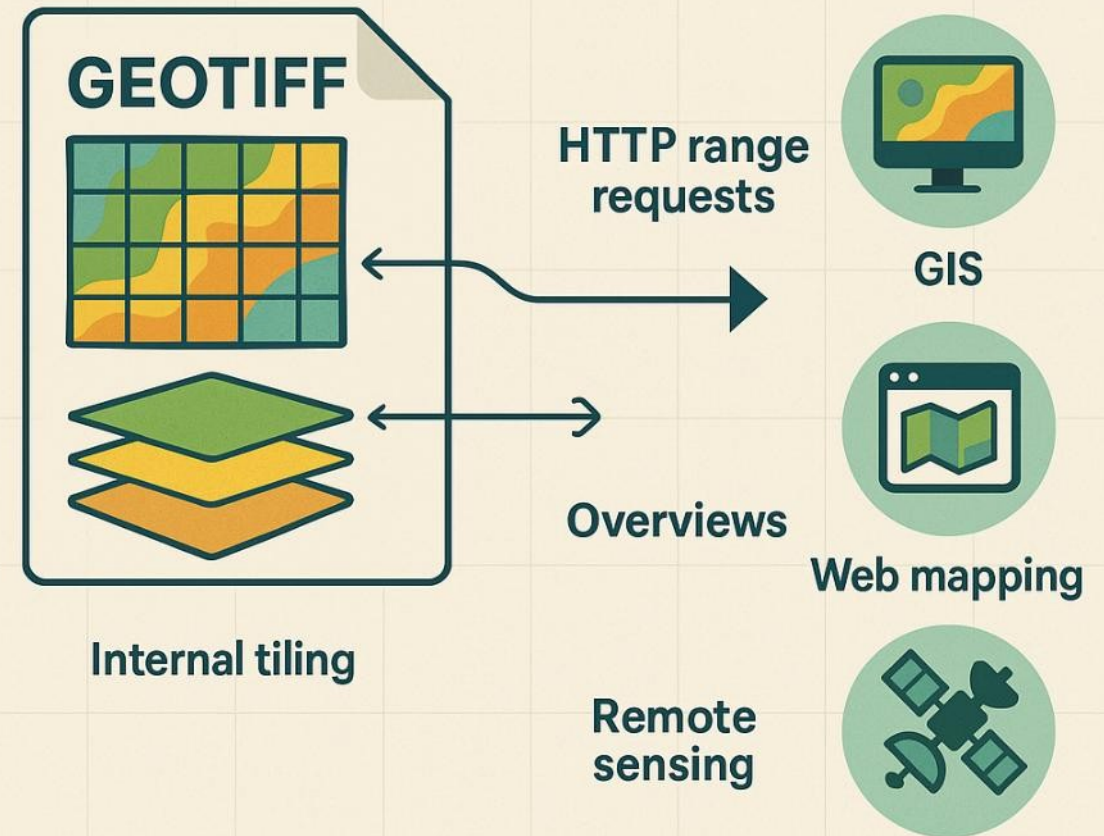
CLOUD TOOL
COMPATIBILITY

Cloud-Optimised GeoTIFF (COG / COTIF)

- A **GeoTIFF** format optimised for cloud access.
- Stores **geospatial raster** data with internal tiling and overviews.
- Allows **HTTP range requests** for partial reads.
- Ideal for **web mapping, remote sensing, and GIS applications**.
- **Supported by tools** like GDAL, QGIS, and web map viewers.

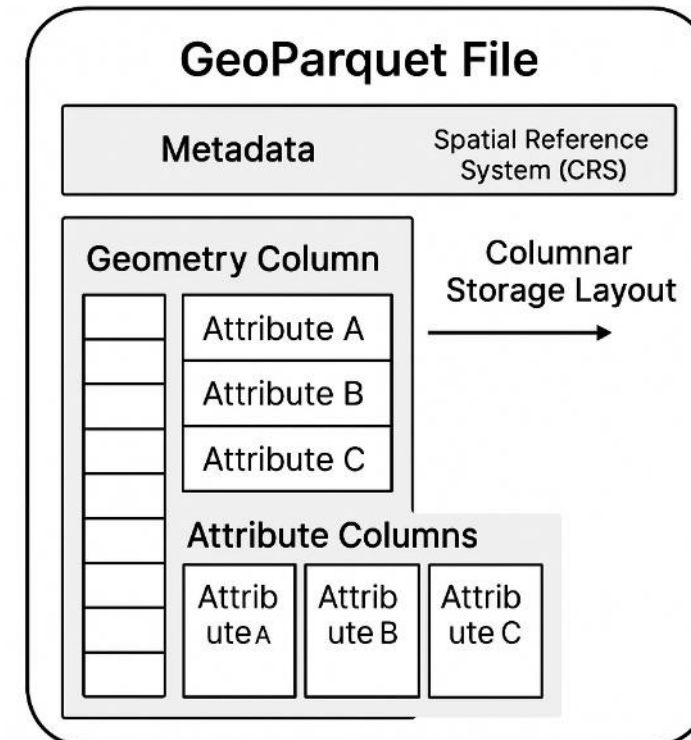
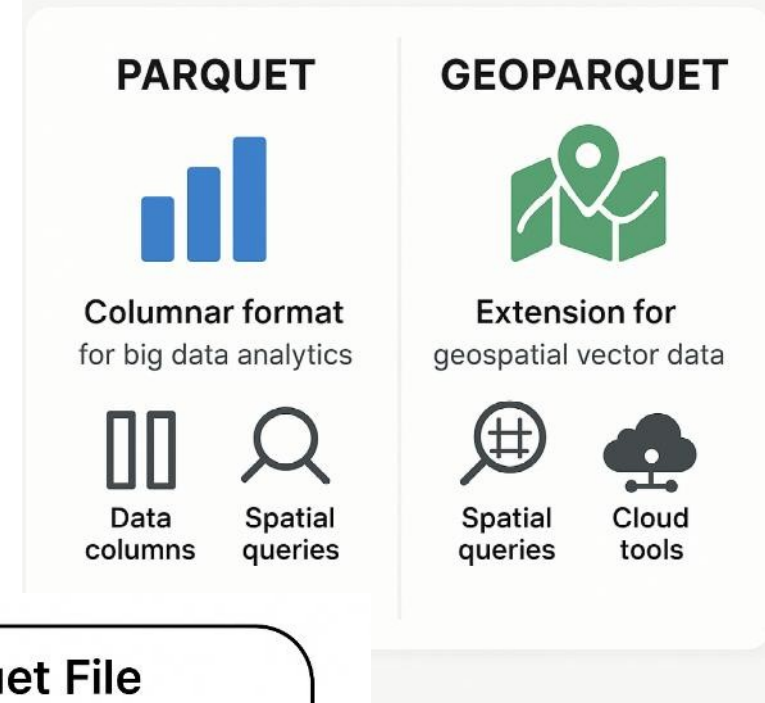


Cloud-Optimised GeoTIFF (COG or COTIF)



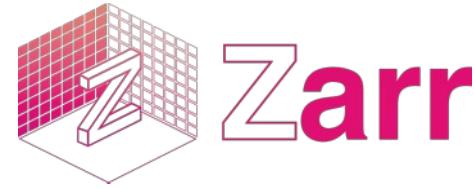
Parquet and GeoParquet

- **Parquet:** A columnar storage format for efficient analytics.
- Optimised for **read-heavy workloads** and big data processing.
- **Flexible schema support** and compression.
- **GeoParquet:** Extension of Parquet for geospatial vector data.
- Stores geometries and spatial reference info.
- Enables scalable spatial queries in cloud-native environments.

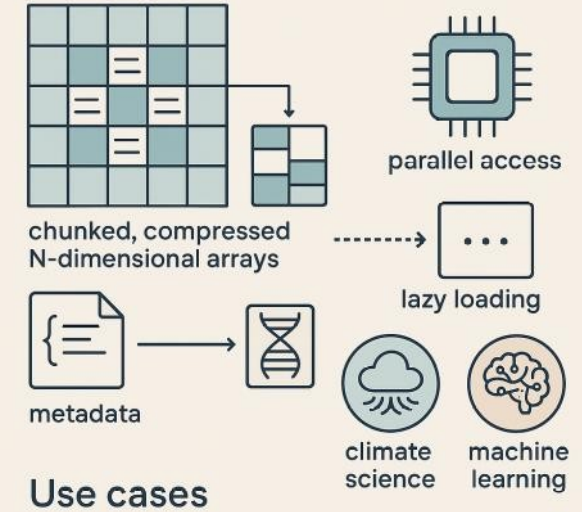


Zarr

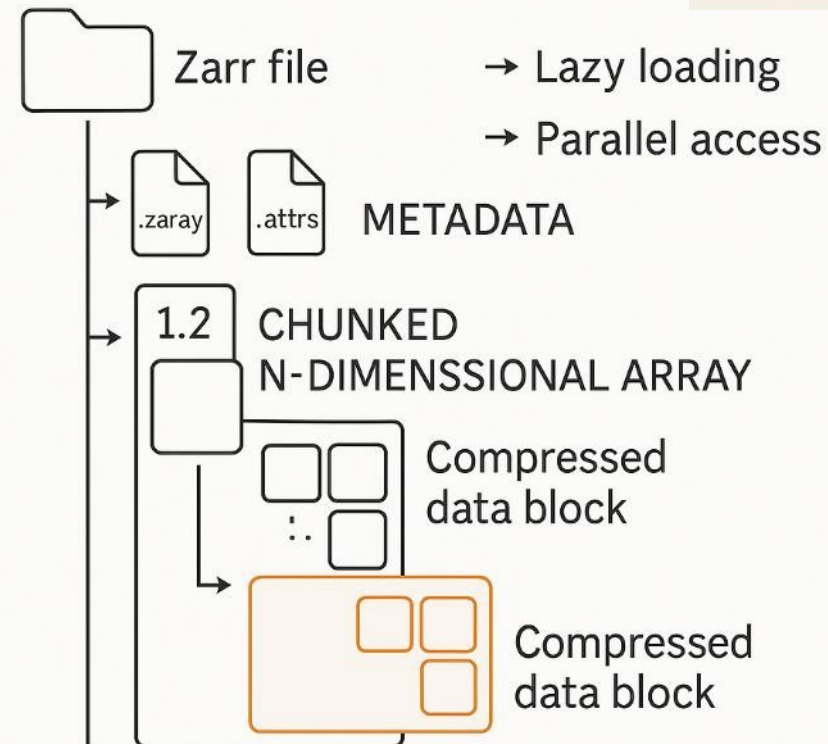
- A format for **chunked, compressed, N-dimensional arrays**.
- Designed for **cloud-native scientific computing**.
- Supports **lazy loading** and **parallel access** to array chunks.
- Common in **climate science, genomics, and machine learning**.
- Compatible with **Python libraries** like xarray and Dask.



Zarr File Format



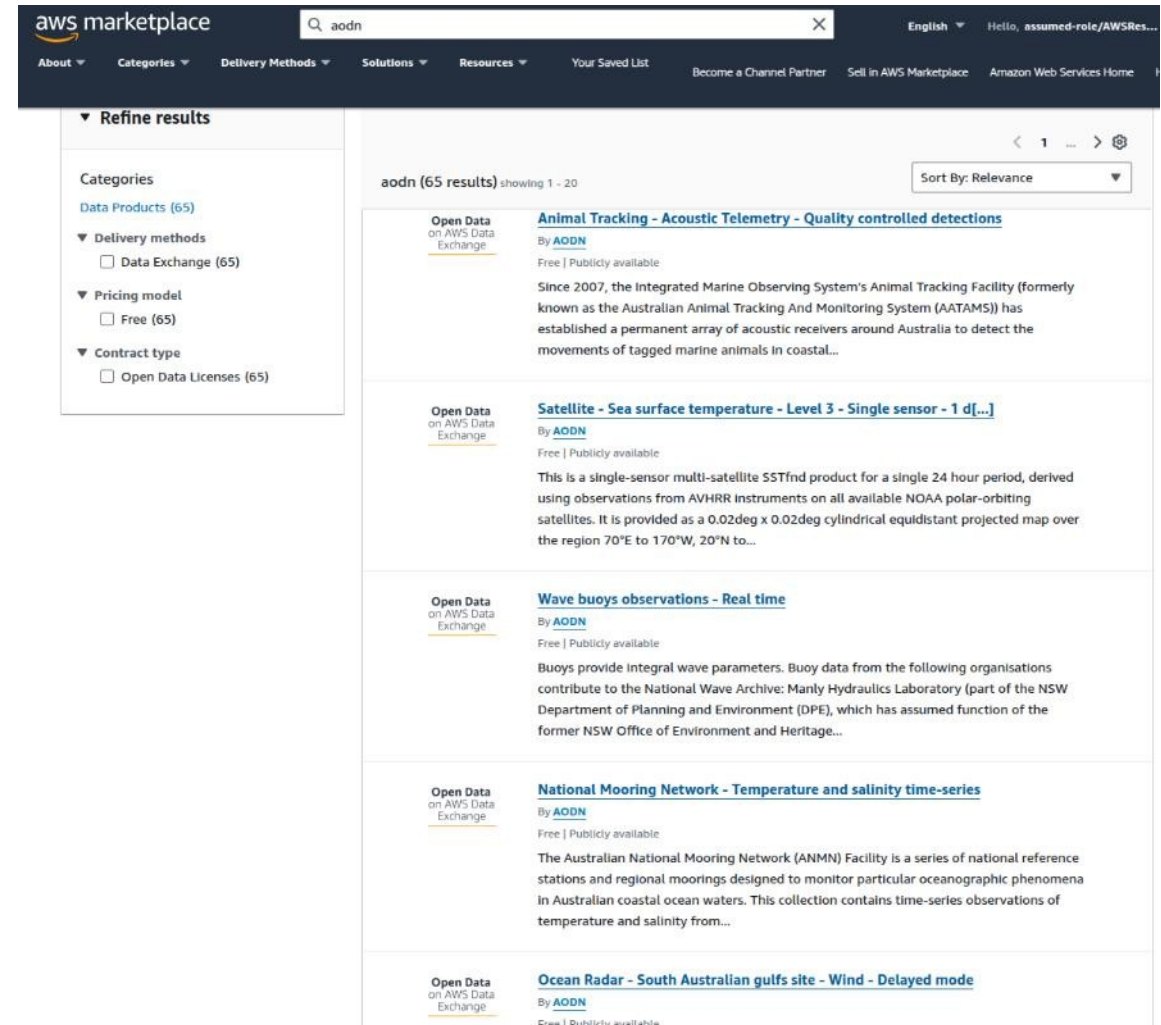
A Zarr file



AODN Cloud Optimised Collection

- Plan for converting all **AODN datasets** into a Cloud Optimised collection (currently 50%)
- All products **available at AWS Open Data Program**
- **AODN Library** for creating and working with CO files
- Every dataset has a (Python) **Notebook**
- At the moment **70 datasets** have been converted to CO format

https://github.com/aodn/aodn_cloud_optimised



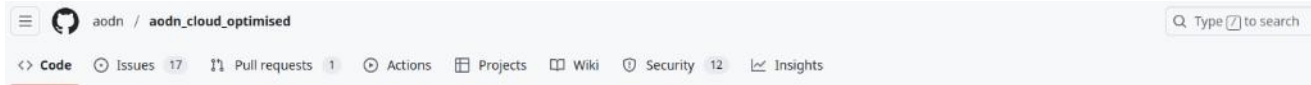
The screenshot shows the AWS Marketplace interface with a search for 'aodn'. The left sidebar shows filters for Categories (Data Products: 65), Delivery methods (Data Exchange: 65), Pricing model (Free: 65), and Contract type (Open Data Licenses: 65). The main results area shows 65 results for 'aodn', sorted by Relevance. The first five results are listed below:

Product Name	By	Availability	Description
Animal Tracking - Acoustic Telemetry - Quality controlled detections	AODN	Free Publicly available	Since 2007, the Integrated Marine Observing System's Animal Tracking Facility (formerly known as the Australian Animal Tracking And Monitoring System (AATAMS)) has established a permanent array of acoustic receivers around Australia to detect the movements of tagged marine animals in coastal...
Satellite - Sea surface temperature - Level 3 - Single sensor - 1 d[...]	AODN	Free Publicly available	This is a single-sensor multi-satellite SSTnd product for a single 24 hour period, derived using observations from AVHRR instruments on all available NOAA polar-orbiting satellites. It is provided as a 0.02deg x 0.02deg cylindrical equidistant projected map over the region 70°E to 170°W, 20°N to...
Wave buoys observations - Real time	AODN	Free Publicly available	Buoys provide integral wave parameters. Buoy data from the following organisations contribute to the National Wave Archive: Manly Hydraulics Laboratory (part of the NSW Department of Planning and Environment (DPE), which has assumed function of the former NSW Office of Environment and Heritage...
National Mooring Network - Temperature and salinity time-series	AODN	Free Publicly available	The Australian National Mooring Network (ANMN) Facility is a series of national reference stations and regional moorings designed to monitor particular oceanographic phenomena in Australian coastal ocean waters. This collection contains time-series observations of temperature and salinity from...
Ocean Radar - South Australian gulfs site - Wind - Delayed mode	AODN	Free Publicly available	

Available at AWS Open Data:

<https://aws.amazon.com/marketplace/search/results?searchTerms=open+data>

AODN Cloud Optimised Collection



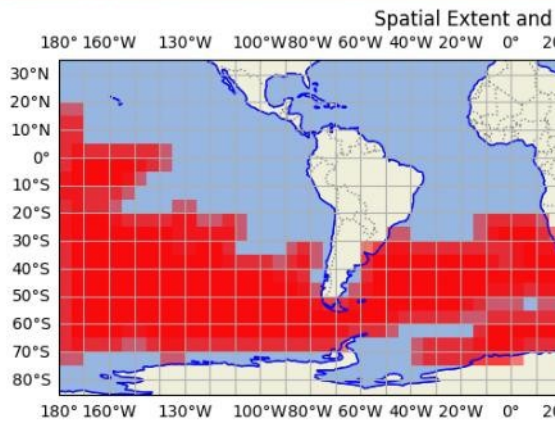
Out[6]: 1145

Visualise Spatial Extent of the dataset

This section plots the polygons representing the areas where data is available. It helps to identify and create a bounding box around the regions containing data.

```
In [7]: aodn_dataset.plot_spatial_extent()
```

/home/lbesnard/miniforge3/envs/AodnCloudOptimised/lib/python3.12/site-packages/cartopy/mpl/feature_artist.py:144: UserWarning: facecolor will have no effect as it has been defined as "never".
warnings.warn('facecolor will have no effect as it has been defined as "never".')



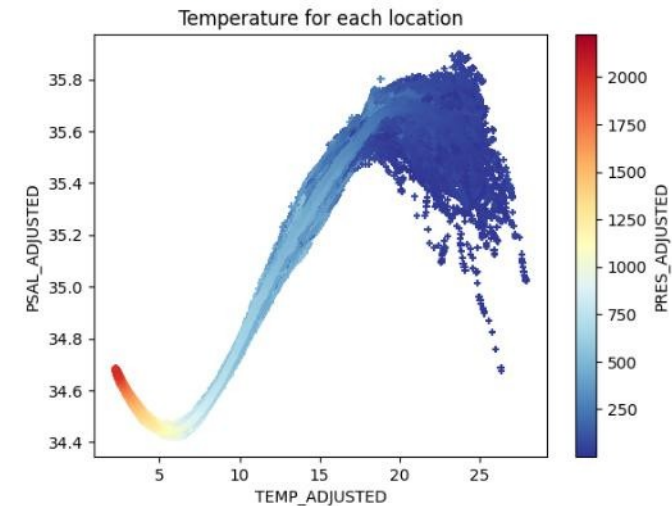
Get Temporal Extent of the dataset

Similarly to the spatial extent, we're retrieving the minimum and maximum values of the TIME values, as the time values are not necessarily accurately representative of the TIME values, as the time values are not necessarily accurately representative of the TIME values.

```
df.aodn.download_as_csv()
```

```
In [11]: df.plot.scatter(x='TEMP_ADJUSTED', y='PSAL_ADJUSTED', c='PRES_ADJUSTED', marker='+', linestyle="None", cmap='RdYlBu')
```

Out[11]: <Axes: title={'center': 'Temperature for each location'}, xlabel='TEMP_ADJUSTED', ylabel='PSAL_ADJUSTED'>



AODN Cloud Optimised Collection

Open Data on AWS Data Exchange

Satellite - Sea surface temperatur

Provided by: AODN, part of the Open D

This product is part of the Open Data and cor
is required. Unless specifically stated in the a
provided and maintained by AWS.

Description

Resources on AWS

Description

This is a single-sensor multi-satellite SSTfnd product for a single 24 hour period orbiting satellites. It is provided as a 0.02deg x 0.02deg cylindrical equidistant p 24 hour average of all the highest available quality SSTs that overlap with that c [data-product-information/](#) [\[2\]](#) indicates where this product fits within the GHRS degC to the SSTskin observations after rejecting observations with low surface v <http://www.bom.gov.au/amoj/docs/2011/beggs.pdf> [\[3\]](#). Matchups with buoy S deviations of 0.6 degC. Refer to the IMOS SST products web page at http://imos.org.au/sstdata_references.html [\[4\]](#) for further information.

License

<http://creativecommons.org/licenses/by/4.0/> [\[5\]](#)

Documentation

<https://catalogue-imos.aodn.org.au/geonetwork/srv/eng/catalog.search#/meta>

How to cite

Satellite - Sea surface temperature - Level 3 - Single sensor - 1 day - Day and ni
https://registry.opendata.aws/aodn_satellite_ghrsst_l3s_1day_daynighttime_slr
The citation in a list of references is: "IMOS [year-of-data-download], [Title], [da

Update frequency

As Needed

Resources on AWS

Description

Cloud Optimised AODN dataset of IMOS - SRS - SST - L3S - Single Sensor - 1 day - day and night time - Australia

Resource type

S3 Bucket

Amazon Resource Name (ARN)

```
arn:aws:s3:::aodn-cloud-optimised/satellite_ghrsst_l3s_1day_daynighttime_single_sensor_australia.zarr
```

AWS Region

ap-southeast-2

AWS CLI Access (No AWS account required)

```
aws s3 ls --no-sign-request s3://aodn-cloud-optimised/satellite_ghrsst_l3s_1day_daynighttime_single_sensor_australia.zarr/
```

Usage examples

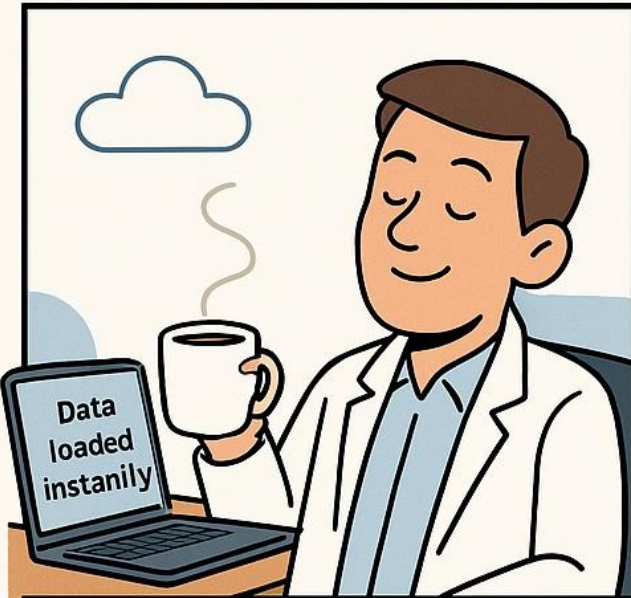
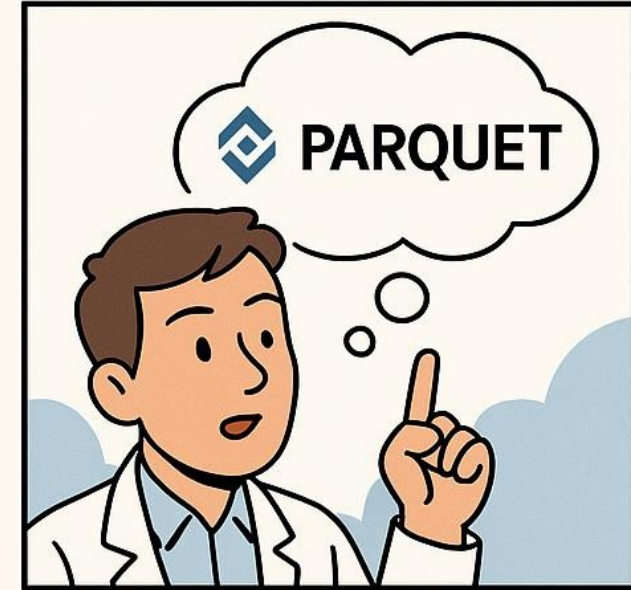
Tutorials

- [Accessing and search for any AODN dataset](#) [\[6\]](#) by Laurent Besnard [\[7\]](#)
- [Accessing IMOS - SRS - SST - L3S - Single Sensor - 1 day - day and night time - Australia](#) [\[8\]](#) by Laurent Besnard [\[9\]](#)

Final remarks

- Cloud-optimised formats are essential for **scalable, efficient data handling** in modern cloud environments.
- They **enable faster access, lower costs, and better performance** across analytics, geospatial, and scientific domains.
- Formats like COG, Parquet, GeoParquet, and Zarr are **tailored to specific data types** and use cases.
- Choosing the **right format** depends on your **data structure**, workflow, and cloud platform.
- Embracing these formats helps **unlock the full potential of cloud-native data** processing.

Cloud-Optimised File Formats



Australia's Integrated Marine Observing System is enabled by the National Collaborative Research Infrastructure Strategy (NCRIS). It is operated by a consortium of institutions as an unincorporated joint venture, with the University of Tasmania as Lead Agent.

PRINCIPAL PARTICIPANTS



SIMS is a partnership involving four universities

ASSOCIATE PARTICIPANTS



IMOS thanks the many other organisations who partner with IMOS, providing co-investment, funding and operational support, including investment from the Tasmanian and Western Australian Governments.

Some of the content has been AI-generated

